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Successful application of predictive information in deep reinforcement learning control: A case study based on an office building HVAC system

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ABSTRACT

Reinforcement Learning (RL), a promising algorithm for the operational control of Heating, Ventilation, and Air Conditioning (HVAC) systems, has garnered considerable attention and applications. However, traditional RL algorithms typically do not incorporate predictive information for future scenarios, and only a limited number of studies have examined the enhancement and impact of predictive information on RL algorithms. To address the issue of coupling RL and predictive information in HVAC system operation optimization, we employed an open-source framework to examine the impact of various predictive information strategies on RL outcomes. We propose a joint gated recurrent unit (GRU)-RL algorithm to handle situations where a time-series exists in state space. The results from four classic test cases demonstrate that the proposed GRU-RL method can reduce operating costs by approximately 14.5% and increase comfort performance by 88.4% in indoor comfort control and cost-management tasks. Moreover, the GRU-RL method outperformed the conventional DRL method and was merely augmented with prediction information. In indoor temperature regulation, the GRU-RL algorithm improves control efficacy by 14.2% compared to models without predictive information and offers an approximately 5% improvement over traditional network models. Finally, all models were made open source for easy replication and further research.

1. Introduction

1.1. Background

Most individuals spend approximately 80%–90% of their lives within buildings [1]. At the same time, buildings are responsible for approximately one-third of global energy consumption and a quarter of CO2 emissions [2,3]. Notably, in building facilities, Heating, Ventilation, and Air Conditioning (HVAC) systems are substantial energy consumers, accounting for 38% of the overall energy usage, which is equivalent to 12% of the final energy consumption [2,4]. As global warming continues to be a growing concern, the energy needs of HVAC systems are predicted to rise even more [5,6].

Well-designed and efficient HVAC system controls play a crucial role in providing occupants with a healthy and comfortable indoor environment while effectively managing energy consumption and carbon emissions. However, the complexity of HVAC system control has increased significantly in recent years. Modern HVAC systems must accommodate various factors, such as integrating on-site renewable energy sources, incorporating energy storage solutions, and responding to grid signals for load shifting to enhance grid stability and security [7]. In contrast to the need for advanced control strategies, the most conventional approach for HVAC system control is rule-based feedback control. This method relies on predetermined schedules to set parameters such as temperature setpoints and utilizes setpoint tracking algorithms such as proportional–integral–derivative (PID) to regulate the system. Although rule-based control (RBC) is simple and effective in maintaining a comfortable range to achieve occupant comfort, it has limitations in considering exogenous factors and predictive information, such as occupancy, electricity price, weather conditions, and demand response signals [8,9]. Consequently, RBC has proven to be a suboptimal approach [10]. Moreover, tuning the gains of PID controllers can be cumbersome, and the control performance may degrade or lead to system oscillation or sluggishness when the operational conditions deviate significantly from those during controller tuning [11,12]. Considerable research attention has shifted towards developing optimal control strategies, particularly those based on machine learning (ML) methods owing to the need to enhance the HVAC system control performance. The advent of big data, powerful computing capabilities, and advancements in algorithms have enabled the exploration and

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Table 1
Summary of studies applying DRL in building HVAC system control.

Ref.	Target system	Control action	Optimization objective	Algorithm	Training environment
[18]	Boiler heating system	Supply water temperature setpoint	Energy consumption; indoor air temperature	DQN	EnergyPlus
[19]	Chiller plant	Chilled water temperature	Energy consumption; thermal comfort	DQN	EnergyPlus
[31]	Heat pump	Heat pump signal	Operation cost; thermal comfort	DDQN; DDPG; SAC	BOPTEST
[20]	Radiant heating system	Supply water temperature setpoint	Energy consumption; thermal comfort	A3C	EnergyPlus
[21]	Air conditioning units	Zone temperature; ventilation fan	Energy consumption; air quality;	DQN	EnergyPlus
[22]	Heating system	Thermostat	Energy consumption; thermal comfort	DQN	MATLAB
[23]	VAV system	Temperature set; fan mass flow	Energy consumption; indoor air temperature	SAC; TD3; PPO; TRPO	EnergyPlus
[24]	Smart HVAC	Power usage; air volume	Energy cost; indoor air quality	DDQN-PRE	Measured data
[25]	VAV	Air temperature; supply water setpoint	Energy consumption; indoor air quality	DQN	EnergyPlus
[26]	Building energy system	Operating ratio	Off-grid operation	DDPG;TD3	Measured data
[27]	Thermal storage	Operation of the cold-water storage tank	Cost	SAC	EnergyPlus

implementation of ML-based control approaches for building HVAC systems [13].

1.2. Reinforcement learning for HVAC system control

Reinforcement Learning (RL), a prominent category of machine learning alongside unsupervised and supervised learning, stands apart because of its unique learning paradigm based on dynamic data acquired during the learning process [14]. The primary objective of RL is to learn the optimal actions that lead to a predefined goal. Through rewards, this reinforcement guides the agent to honor its decision-making abilities and achieve the desired objective [10,15].

Deep Reinforcement Learning (DRL) shares the foundational principles of classical RL; however, it distinguishes itself by employing complex deep neural networks (DNNs) as RL agents instead of simple tabular settings or linear functions. Leveraging DNNs endows the RL agent with a higher representational capability, enabling adaptation to intricate control problems and facilitating the resolution of real-world complexities that demand a large number of states and actions to properly represent the control problem [16]. DRL is a highly suitable approach for HVAC system control since building HVAC systems is non-linear, time-varying, and uncertain complex [17]. Many studies have applied DRL techniques to various aspects of HVAC system control, including the primary subsystem [18,19], secondary subsystem [20–23], entire HVAC system [24,25], and associated energy management system [26–28]. An overview of studies applying DRL to HVAC system control is presented in Table 1. These advanced control strategies have shown remarkable potential for HVAC system control, significantly improving indoor comfort conditions while simultaneously reducing energy consumption. However, the DRL process involves testing and evaluating new strategies through trial and error and subsequently using these evaluations to refine its control policy. This experimental approach poses challenges for the implementation of DRL controllers in real-world buildings. To address this limitation, an effective method involves pretraining the DRL controller offline by utilizing a surrogate virtual environment [27]. Many studies have developed customized HVAC system models based on simulation platforms such as EnergyPlus and MATLAB, enabling the DRL controller to interact within these virtual environments [10]. Nevertheless, accurately modeling the full spectrum of real-world environmental features is challenging, potentially leading to poor generalizability of the controller when applied to actual buildings [29,30]. Furthermore, the influence of future information, such as weather conditions, significantly influences the performance of HVAC system control, particularly regarding precooling and preheating operations [10]. To address this issue, an increasingly promising approach involves integrating predictive information into a DRL controller.

1.3. Predictive model integrated with RL for HVAC system control

Recurrent Neural Networks (RNNs) are a type of Artificial Neural Network (ANN) specifically designed to address tasks like language modeling, machine translation, and speech recognition. Their defining feature lies in their recurrent connections, which enable them to maintain a hidden state that carries information from previous time steps, thereby influencing the current step. This characteristic makes RNNs suitable for processing sequential data such as natural language and time-series sequences. Considering that data related to buildings inherently involve time-series data and that building performance is affected by short-term and long-term variations, such as occupancy and seasonal changes, intrinsic temporal dependencies are prevalent [32]. As a modified version of RNN, Long Short-Term Memory (LSTM) addresses the vanishing gradient problem and long-term dependency issues associated with traditional RNNs [33]. Instead of using a recurrent layer, the LSTM recurrent unit processes information by employing various gates (forget and input gates) to control the information that should be remembered, forgotten, or updated in the cell state. Subsequently, it produces a new hidden state that passes through the output gate for use in the subsequent time step. Given this perspective, LSTM has found widespread application in the field of building information prediction, including control performance, energy consumption, thermal comfort, and air quality [34–37].

For successful integration with a DRL controller, the predictive model must be capable of offering continuous feedback for control operations, which makes the RNN model (specifically, LSTM) a suitable candidate [29]. Despite its potential to improve control performance, the application of integrating predictive information into optimal HVAC control using DRL remains limited in current literature. Zou et al. [38] utilized LSTM in combination with a DDPG agent to control air handling units (AHU), achieving 27%–30% energy savings while ensuring thermal comfort. LSTM models utilize historical Building Automation System (BAS) data, approximate HVAC operations, consider the current state and action as inputs, and provide the next state and reward as outputs. Pinto et al. [39] employed an LSTM coupled with an SAC agent to optimize heat pumps, chilled water, and domestic hot water storage across four buildings. Their approach effectively reduced electricity consumption costs by 23%. Blad et al. [40] addressed the issue of poor behavior during the early training of an RL algorithm by coupling LSTM with a Q-learning-based multi-agent algorithm for the online fine-tuning of HVAC systems. As a result, they successfully reduced heating costs by 19.4% compared to traditional control policies. Zhuang et al. [29] proposed a set of 16 LSTM-based model architectures that incorporated different combinations of convolutional neural networks (CNN), bidirectional processing, and attention mechanisms (AM) to identify the best-performing model setups. The findings indicate that the attention mechanism notably enhances the prediction performance in both recursive and independent prediction scenarios. These optimal models were then integrated with an SAC RL agent to analyze sensor metadata and optimize HVAC operations, resulting in significant energy savings of 17.4% and a remarkable 16.9% improvement in thermal comfort within the surrogate environment. Table 2 summarizes the current studies that integrate predictive models with RL for HVAC control. However, further research is required to explore and harness the potential of integrated controllers in this domain.

Table 2
Summary of studies integrated predictive model with DRL.

Ref.	Target system	Optimization objective	Algorithm	Predictive model
[38]	AHU	Energy consumption; thermal comfort	DDPG	LSTM
[39]	Energy storage	Energy consumption; thermal comfort	SAC	LSTM
[40]	Heat pump	Energy cost; indoor air quality	Q-learning	LSTM
[29]	AHU	Energy consumption; PMV values	SAC	LSTM

1.4. Objective of this research

This study aims to address the limitations of current predictive models integrated with reinforcement learning for HVAC system control. Through a comprehensive review and discussion, the authors identified the following limitations of the existing approaches:

- Lack of Reproducibility and Comparison: Many studies use customized virtual simulation models as the training environment for DRL controllers. This approach hinders the reproducibility and testing of developed algorithms, making it challenging to compare different control algorithms for specific applications. Additionally, the use of simulation platforms like EnergyPlus and MATLAB, which do not fully model the dynamics of HVAC systems and sometimes use idealized controllers [41], may limit the generalizability of the developed algorithms to real buildings.
- Table 2 summarizes the studies combining Reinforcement Learning (RL) and forecasting information in HVAC systems. Most of these studies incorporate networks such as LSTM for future predictions and then feed the predictive data into RL. The number of studies is relatively small, indicating that the research is not yet comprehensive. The discussion has not yet touched on how to correctly utilize forecasted data in scenarios where such information already exists. This situation is quite prevalent in optimizing HVAC systems, for instance, using weather forecasts for anticipated outdoor conditions.

Considering these limitations, this study focuses on exploring the effective application of predictive data in reinforcement learning. We compare traditional networks and RNN in terms of their ability to handle predictive information from observational data. Simultaneously, we explored the potential impact of this predictive information on the optimization of the HVAC system operation. The presented work encompasses the development of an open-source test platform and algorithms using the BOPTEST platform [42] and Modelica-based models, focusing on the dynamic behavior of HVAC systems to enhance reproducibility. It also explores the integration of RL and predictive information in optimizing HVAC operations, evaluating the efficacy of different approaches in handling predictive data. The proposed combination of RNN and RL demonstrates superior performance over conventional neural network approaches in both tasks, establishing a valuable reference for future research in this field.

2. Methods

2.1. Emulator environment

To fully verify the effect of the proposed reinforcement learning algorithm, we used a building optimization testing (BOPTEST) framework as a benchmark [42]. BOPTEST is an open-source standardized building simulator that includes several types of buildings and HVAC systems. The open-source framework improved the transparency of the testing algorithm, and the reproducibility of the framework was very strong.

A conceptual diagram of BOPTEST is illustrated in Fig. 1. All BOPTEST's building models are developed using Modelica and exported as Functional Model Units (FMUs). These are bundled into Docker containers complemented by boundary conditions and external disturbances such as a year's worth of weather data. The application programming interface (API) allows an external controller to access observed and forecasted data, modify control signals, choose test scenarios, establish simulation environments, and proceed with the simulation. BOPTEST provides a standardized set of key performance indicators (KPIs) for objective performance assessments, incorporating factors such as energy consumption, thermal discomfort, operational expenses, indoor air quality discomfort, emissions, and proportion of computing time.

2.2. Basic definitions and methods of RL

Before discussing the algorithms, we first established a foundational understanding of reinforcement learning and its fundamental parameters. An in-depth explanation is provided in Ref. [43]; here we provide a concise overview to ensure the coherence of the article.

RL engages in a process of learning via trial and error, directing behavior based on rewards achieved from environmental interactions, aiming to maximize cumulative rewards. Each RL challenge can be represented using a Markov Decision Process (MDP) [44]. The Markov Decision Process (MDP), characterized by quintuples (S, A, P, R, γ), is employed for scheduling optimization within a building renewable energy system.

- S: Reinforcement learning agents utilize accessible information, encompassing current, historical, and predictive data, to inform their decisions. For instance, in managing a building's HVAC system temperature, an RL agent bases its decisions on factors such as indoor and outdoor temperatures, solar radiation intensity, and future temperature forecasts.
- A: Actions reflect the variety of decisions an agent makes in response to diverse environments; for instance, factors like airflow and air temperature.
- P: State transition probability depicts the state's distribution at the subsequent time step given a certain action in a particular state. If the RL algorithm learns an explicit state transition probability, it is considered model-based. Its counterpart is referred to as model-free RL [26].
- R: The reward function establishes the computation of the immediate reward achievable for a specific state–action pair. This reward calculation could be a defined value or a mathematical expectation. The reward magnitude directs the optimization course.
- γ: In RL problems, due to feedback delays, the current step's action might require several steps to take effect. Therefore, the agent should not merely select the action yielding the highest immediate reward but should anticipate the outcomes of future steps. The discount factor, γ, enables RL to equilibrate short- and long-term objectives. By discounting future rewards, we can ascertain the value of the agent's present actions, which equates to the cumulative future rewards and signifies the true optimization target of RL.

The foregoing delineates the core elements and enhancement objectives of RL. Conventional RL approaches typically employ tables to manage finite actions and states. This strategy is effective for simplified states and offers a fast convergence. However, in real-world scenarios, these discrete states and actions are often reduced. The exterior

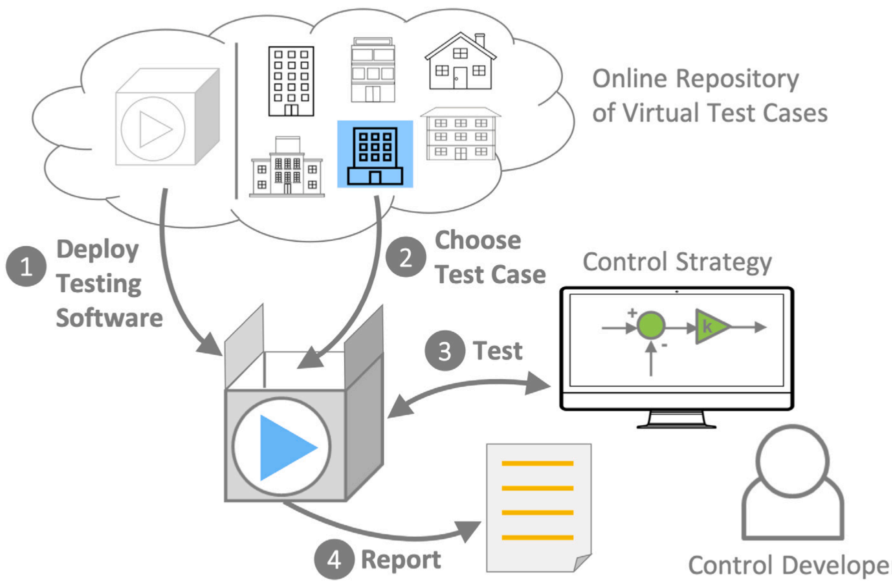

Fig. 1. Conceptual framework of BOPTEST [42].

temperature of a building is an example of a typical continuous-state value. DRL and RL share the same theoretical underpinnings. This unique distinction lies in DRL's utilization of a DNN instead of a conventional table to approximate cumulative rewards, offering more effective management of continuous vector spaces. This study primarily focuses on research involving DRL.

2.3. DRL algorithms

Deep reinforcement learning agents require deep neural networks for function approximation to handle expansive state spaces. The deep Q network (DQN) algorithm made significant strides in 2016, outperforming humans in the Atari games [16]. Nevertheless, the DQN's inability to effectively manage the RL problem in a continuous action space presents a considerable limitation to its further progression and widespread application. The Deep deterministic policy gradient (DDPG) algorithm, which employs an actor–critic (AC) strategy and a deterministic policy gradient (DPG), refines the DQN algorithm for application in continuous action spaces [45].

2.3.1. Actor–critic method

In a DQN, the algorithm lacks a distinct actor; instead, it opts for action selection via a greedy strategy. However, this approach falls short of continuous action spaces, rendering the DQN algorithm inapplicable to such contexts.

Compared with the DQN, the Actor–Critic (AC) approach enables reinforcement learning in a continuous action space via a standalone actor network [46,47]. The AC model comprises two networks: an actor network μ (parameterized by θ^{μ}) and a critic network Q (parameterized by θ^{Q}). The network configurations of the actor and the critic may differ depending on the algorithm used. Critic network Q is tasked with evaluating the current action and computing the action-value function. The actor network is updated based on the feedback from the critic network, and it uses the input state vector to derive the current action. This action is computed by maximizing the value of the critic network.

In the Actor–Critic (AC) framework, the Q network update approach employs the Bellman equation, as shown in Eq. (1).

$$\Delta Q_t = r_t + \gamma \bullet Q\left(s_{t+1}, \mu\left(s_{t+1} \mid \theta^{\mu}\right) \mid \theta^{Q}\right) - Q\left(s_t, a_t \mid \theta^{Q}\right) \tag{1}$$

In Eq. (1), the sum of the current immediate reward r_t and the discounted future action-value function serves as our update target. By subtracting the present action-value function, we obtain the updated gradient of the Q network.

We present an update to the actor network in Eq. (2), following [48]. The computation formula only involves gradients with respect to the actor network parameters. This approach enables the actor network to produce continuous actions and discover actions that optimize the Q network's output.

$$\nabla_{\theta^{\mu}} \mu = \nabla_a Q\left(s, a \mid \theta^{Q}\right)\Big|\, s = st, a = \mu\left(s_t\right) \nabla_{\theta_{\mu}} \mu\left(s \mid \theta^{\mu}\right)\Big|\, s = st \tag{2}$$

2.3.2. Selected algorithms

DDPG leverages the principles of AC and DQN to construct actor networks by introducing a reinforcement learning algorithm capable of handling continuous action spaces. However, the update process of the DDPG algorithm has many problems with unstable training, and it is easy to enter a local optimal solution prematurely. As an improved model of DDPG, Soft Actor–Critic (SAC) uses the maximum entropy model to fully improve the exploration efficiency of the model so that the model will not fall into the local optimal solution in the later stage of training [49]. This study examines the performance of this algorithm, which follows an actor–critic paradigm, namely, SAC. SAC employs entropy regularization to address the inefficiency encountered during the later stages of DDPG training. The entropy regularization's goal is to maximize randomness in the strategy to avoid stagnation at local optima while simultaneously optimizing the expected returns.

We selected SAC based on the following considerations:

- Our HVAC control problem features a continuous action space. SAC algorithms are inherently adept at managing both continuous state and action spaces.
- SAC performs well in previous work on building energy management. Brandi et al. employed an SAC agent to control a heating system, resulting in 2%–6% energy conservation and up to a 65% increase in thermal comfort [50]. Wang and colleagues also evaluated the performance of SAC and DDPG within the BOPTEST framework [31].
- In this study, the authors focus on the impact of introducing predictive data on RL algorithms and how to better process predictive information. Then the introduction of prediction information should take effect for all RL algorithms. Therefore, the authors also selected SAC to verify the validity of the predicted data.

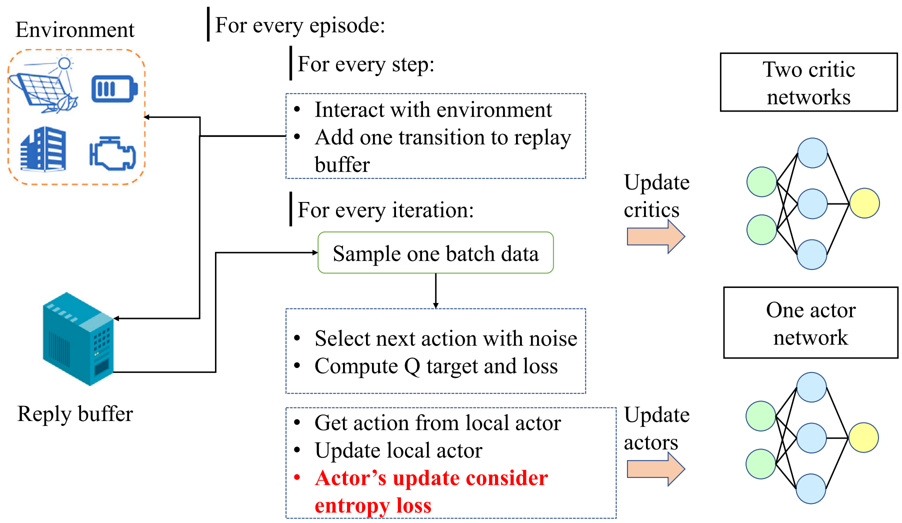

Fig. 2. Training procedures of Soft Actor–Critic (SAC).

Fig. 2 presents a flow diagram of the SAC algorithm, which is a variant of the Actor–critic algorithm family. SAC distinguishes itself by incorporating the concept of entropy to enhance policy exploration. This approach helps in avoiding local optima, as indicated by the red text in the diagram.

2.4. How to combine predictive information with RL algorithms?

This study focused on investigating how incorporating prediction information can enhance and optimize reinforcement learning algorithms. From an algorithmic perspective, the incorporation of predictive data should primarily consider alterations to the state vector, meaning that predictive information is integrated directly into the state vector itself. In the context of HVAC systems, future information, which is frequently presented as time-series data such as outdoor air temperature and solar radiation fluctuations over the next 24 h, is often vital. Hence, employing an RNN to process predictive data using the DRL algorithm is intuitive. Some researchers considered combining DDPG and RNN networks [51]. Therefore, we discuss the influence of the combination of the DRL algorithm and RNN on the final operation control effect for HVAC system operation control.

Fig. 3 shows how we introduce prediction information and use RNN to process the prediction information. Given that SAC utilizes AC frameworks with distinct actor and critic networks, we incorporated an RNN layer at their base. This enabled the processing of sequential data for prediction information.

In essence, we integrated forecast data and current observations to form a multivariate time-series, which we subsequently processed using the RNN network. For Actor and Critic networks, we simply regard the RNN network as a feature extractor, hoping to obtain more information from the features of the time-series to help in decision-making.

In this study, we employed a Gated Recurrent Unit (GRU) layer to process the observations for the RL algorithm. In essence, the computations involved in the GRU are parallel to those in LSTM [52]; however, the GRU streamlines the gating mechanism, thereby simplifying the calculation process. The GRU performs subsequent functions for each component of the input series (Eq. (3)):

$$
\begin{aligned}
r_t &= \sigma\left(W_{ir}x_t + b_{ir} + W_{hr}h_{(t-1)} + b_{hr}\right)\\
z_t &= \sigma\left(W_{iz}x_t + b_{iz} + W_{hz}h_{(t-1)} + b_{hz}\right)\\
n_t &= \tanh\left(W_{in}x_t + b_{in} + r_t * \left(W_{hn}h_{(t-1)} + b_{hn}\right)\right)\\
h_t &= \left(1 - z_t\right) * n_t + z_t * h_{(t-1)}
\end{aligned}
\tag{3}
$$

The term x_t denotes the value at different time steps in the state vector, such as outdoor temperature prediction at time t. $h_{(t-1)}$ represents the hidden layer state computed at the previous time step. The symbols r_t, z_t, and n_t denote reset, update, and new gates, respectively. These computational methods bear a resemblance to the LSTM. For a more comprehensive explanation of the computations and principles underlying the GRU, readers can refer to the official PyTorch documentation [53].

3. Experiment setting

3.1. Building envelope and HVAC system

The study employs 'BESTEST air' as a test subject, which refers to a singular room situated in proximity to Denver, CO, USA, occupying a floor area of 48 m^2. The room configuration included four exterior walls facing the principal directions along with a flat rooftop. A pair of 6 m^2 windows was embedded in the southern wall. The room was primarily utilized as a two-person office, marked by a light load density.

The thermal management system of the office consisted of an idealized four-pipe fan coil unit (FCU) that handled both heating and cooling tasks. This unit included a fan, cooling coil, heating coil, and filter. It operates by drawing room air into the unit, propelling it over the coils, filtering it, and redistributing the conditioned air into the room. A variable-speed drive powered the fan motor. The cooling coil used chilled water from the chiller, whereas the heating coil used hot water from the gas boiler. The central plant was not included in the model. We manage the HVAC system by modulating the supply air temperature within a range of 12 °C to 40 °C and standardizing the air mass flow rate in proportion to the designated design flow rate, which ranges from 0 to 1. Further details on the case studies can be found on the reference webpage [54].

3.2. Design of state space and ablation study for prediction observations

The emulator model offers an abundance of outputs. The selection of the state space is critical for the performance of the RL controller. The state variable should encapsulate both present and projected states, providing information pertinent to the agent's decision-making process. In addition, acknowledging the possibility that detailed measurements, such as those provided by BOPTEST, might not be available in practical engineering applications, we selected our observations not only considering their potential influence on indoor temperature control but also favoring more commonly available measurements. After several trials, we identified the state space by incorporating the eight variables listed in Table 3. The rationale for choosing these specific state variables is outlined below:

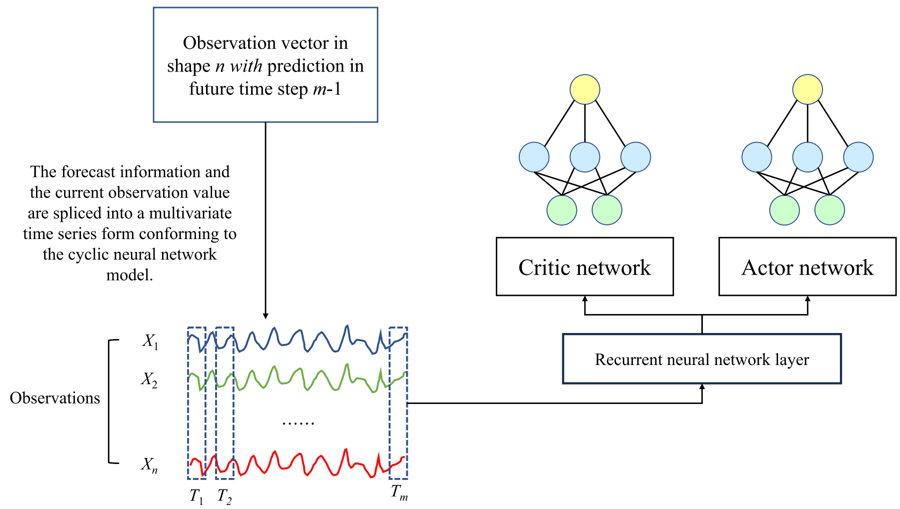

Fig. 3. Combination of Recurrent Neural Network and Reinforcement Learning.

Table 3
State variable description and associated the minimum and maximum value.

State variable [Unit]	Min. Value	Max. Value	Has prediction
Location of the state in episode [1]	0	167	No
Zone air temperature [K]	280	310	No
Direct normal radiation [w/m^2]	0	982.62	Yes
Global horizontal solar irradiation [w/m^2]	0	1027.25	Yes
Outside relative humidity [1]	0	1	Yes
Outside drybulb temperature [K]	248	310	Yes
Wet bulb temperature [K]	248	294	Yes
Sky cover [1]	0	1	Yes
Power price [$]	0.0444	0.13814	Yes
Lower bounds of the comfort range [K]	280	310	Yes
Upper bounds of the comfort range [K]	280	310	Yes

- The states of ‘Location in the Episode’ and ‘Zone Air Temperature’ are based on the relevant features of the current timestep in the HVAC system’s reinforcement learning. It is necessary to know the location of the current timestep within the episode to accelerate convergence. Meanwhile, indoor temperature is one of the control targets. However, predictions cannot be made due to the uncertainty of future actions.
- Outdoor temperature, humidity, and solar radiation parameters can impact the heating and cooling load of HVAC systems, so we have incorporated these parameters into the state vector. Predictions for these variables can usually be sourced from weather forecasts; hence, we utilize these forecasted state vector values in our corresponding models.
- The changes in electricity costs in this study adopt a dynamic mechanism where the cost of system power purchasing varies at different times of the day. The authors anticipate that the RL agent can learn the optimal strategy in the context of dynamically changing electricity prices. Detailed information regarding the variations in electricity prices can be found on the related case web page.
- The comfort temperature range significantly influences the satisfaction of office users and is an essential aspect of this study. Moreover, the variations in the comfort temperature range can also provide information about the current occupancy status of the room, which can be obtained through RL algorithms. Moreover, the authors wish to emphasize that BOPTEST maintains a set temperature for office comfort, specifically between 21 to 24 degrees Celsius during occupancy, and 15 to 30 degrees Celsius when unoccupied, regardless of the season.

3.3. Design of state space and ablation study for prediction observations

We have clarified the reasons for choosing these observations. Most of these data points yielded forecasts for future events. For example, outdoor temperature predictions can be obtained from weather forecasts, while future electricity prices can be obtained directly and accurately. We posit that predictive information can positively contribute to RL optimization and propose an effective method for handling such predictive information. It is important to note that this study directly employs the predictive API provided within the BOPTEST framework, without developing any independent predictive models or algorithms. We designed three distinct methods for processing predictive information to verify the efficacy of GRU. Other configurations of the RL algorithm, along with the random seed, remained consistent to ensure the validity and reproducibility of the ablation studies. The three processing methods and their corresponding observational space dimensions are as follows:

- M.1 (baseline): M.1 will employ the most traditional reinforcement learning algorithm and state space configuration, namely choosing the actual measurements of the current time step as observations without incorporating any forecast information about the future. The performance of the remaining models incorporating predictive information will be compared with M.1, serving to validate the enhancement effect of predictive information on RL algorithms. From Table 3, it can be observed that the state space dimension for M.1 is 11, indicating that all 11 variables only take the value at the current time step.
- M.2 (prediction by flatten): Starting from M.2, we incorporate predictive information into the state space. However, despite introducing predictive information in M.2, we will not alter the Actor and Critic network structures. Instead, we flatten all the information into a column vector and feed it into the reinforcement learning process for training. In Table 3, we have a total of 9 states containing future predictive information. We incorporate the current and future predictive states of these 9 variables into the state space. The two variables without predictive information are included with their current information only. For instance, when the predictive timestep is 23, the dimension of the state vector is $9 * 24 + 2 = 218$.
- M.3 (prediction by GRU): M.3 employs the same predictive information as M.2; however, it utilizes GRU networks in the Actor and Critic to process states. Fig. 4 illustrates the method M.3 employs to process the state space.

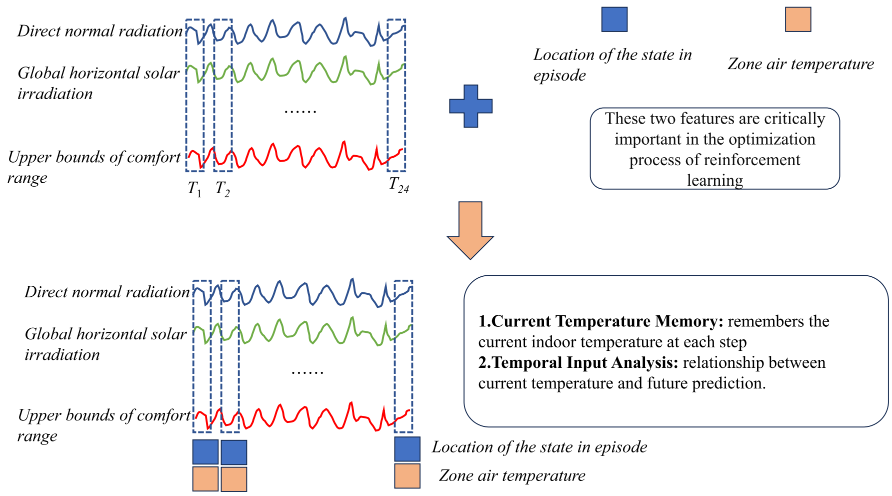

Fig. 4. Observation spaces generation methods for M.3.

Due to the sequential nature of GRU inputs at each time step, we designed the network to remember the current indoor temperature at every step of computing predictive information. This approach emphasizes the significance of the current temperature, preventing it from being overlooked by the network due to its small dimensional representation. Additionally, the input at each step allows the network to consider the relationship between future time steps and the current step, enhancing its ability to make more informed decisions based on a comprehensive understanding of temporal dynamics.

To comply with the input requirements of the GRU network, we duplicate the two values with no predictions across all forecasted time steps. For instance, when the predictive timestep is 23, the dimension of the state vector is $11 * 24 = 264$.

3.4. Design of action space

The indoor temperature and operational costs were regulated using air-conditioning control signals. In this scenario, the decision variables were the supply air temperature and airflow signals, which aim to maintain the building within a comfortable range. The range of the airflow signal spans [0,1], which is a variable normalized in advance within the BOPTEST framework. The range of the supply air temperature was [288.15, 313.15], as measured in Kelvin. In summary, the action space encompasses two dimensions, both of which constitute a continuous action space.

3.5. Design of reward function

The design of the reward function establishes the optimization direction and goals of the RL algorithms. To better evaluate the role of predictive information in RL, we devised two distinct reward functions and optimization objectives.

3.5.1. Reward based on user comfort and running costs

Our objective was to minimize thermal discomfort and operational costs as much as possible. Consequently, the reward function was structured as a weighted sum of these two goals. Thermal discomfort signifies the total deviation of a region's operative temperature from the established comfort zone over a specified time period, expressed in K h. Operational cost represents the expenditure incurred under a given pricing scenario within a specific timeframe, measured in terms of monetary units ($\$/\mathrm{m}^2$). Moreover, within the BOPTEST framework, three distinct pricing models are identified: constant, dynamic, and highly dynamic electricity prices. Here, we used the Dynamic Electricity Price. BOPTEST automatically calculates the operation costs according to the electricity price corresponding to user selection. A mathematical representation of the objective function is given in Eq. (4):

$$J_k = \omega \cdot C + \delta \tag{4}$$

Here, C represents the operational cost, and δ signifies the thermal discomfort. ω serves as a weight parameter, providing a balanced trade-off between operational cost and thermal discomfort; this is another crucial hyperparameter that requires further tuning. The operational cost and thermal discomfort values can be sourced from the BOPTEST's APIs. Considering our aim to optimize the cumulative reward, we can define the immediate reward at a specific time as follows (Eq. (5)):

$$r_{k+1} = -\left(J_{k+1} - J_k\right) \tag{5}$$

With this computational methodology, the maximum value of the reward was zero. A pivotal decision in designing the reward function is to determine the value of the weight ω, which aims to balance the tradeoff between operational cost and thermal comfort. A higher value of ω indicates a preference for reduced operational costs over comfort. Theoretically, ω should be selected according to the preferences of the building occupants. However, a survey on how to choose ω is beyond the scope of this study because the aim is to compare the impact of predictive information on RL algorithms. Moreover, the selection of weights should consider the absolute numerical differences between the two objectives. We assigned a value of 16 to the weight parameter, which can be adjusted for other applications.

3.5.2. Reward based on thermostatic control

In Section 3.5.1, we develop a reward function using BOPTEST's inherent API to fulfill the requirements of both indoor thermal comfort and operating cost reduction. However, the presence of two criteria in the optimization objective is disadvantageous when comparing the degrees of enhancement of the predictive information, and the GRU contributes to the RL algorithm. This is because of the potential improvement in one criterion, whereas the other worsens. In this study, the authors compared another control method for HVAC systems, specifically, thermostat control. The objective of this system was to maintain the indoor temperature at a fixed value. Here, we disregard the operational cost of the system and focus solely on a single objective to facilitate a comparison of the ultimate effects of diverse predictive information on RL. Here, we establish a reward function for constant temperature control by creating a Gaussian distribution. The viability of this reward function was confirmed in prior studies using RL algorithms [26,46].

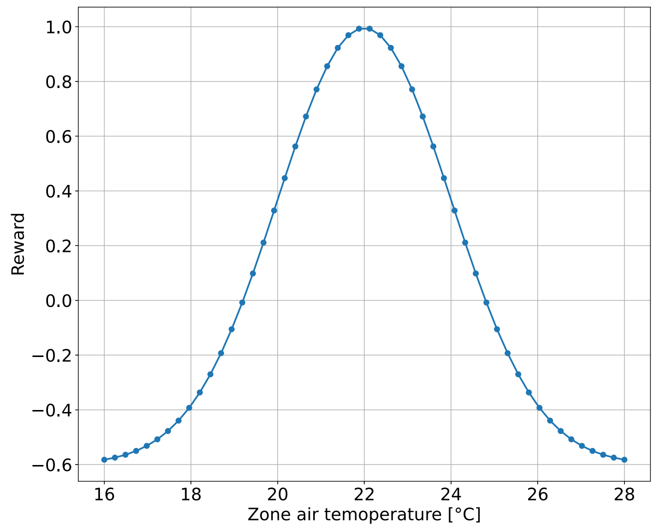

Fig. 5. Structure of thermostatic control reward function.

First, we compute the temperature disparity for the present time interval by deriving the optimization strategy for each stage. Here, we make the assumption that the HVAC system aims to maintain the indoor temperature at 22 degrees Celsius.

Upon completion of the calculations, we were guided by the properties of the Gaussian distribution (Eq. (6)), have set the mean of the Gaussian distribution to 295.15 and the standard deviation to 2 for the purposes of this study. Following this, we scaled the Gaussian distribution to a certain extent, resulting in a maximum value of 1, a minimum value of −0.6, and both positive and negative values. The scaling technique is expressed in Eq. (7).

We obtained the reward function shown in Fig. 5 by leveraging the probability density function of the Gaussian distribution. For ease of display, Celsius is used to plot in this context.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{6}$$

$$f(x) = 8 * \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} - 0.6 \tag{7}$$

where x is the zone air temperature from BOPTEST and π indicates the control target of our optimization, 293.15 K for instance.

This Gaussian distribution transforms the computed error value into a continuous reward value, ranging from −0.6 to 1, an interval ideal for gradient descent. Moreover, the optimization target can be conveniently altered by adjusting the mean of the Gaussian distribution. For instance, setting the mean to 23 degrees Celsius indicates our aim to control the room temperature at 23 degrees Celsius.

3.6. Experimental setting

The target building dataset incorporates hourly data measurements. The model was trained using data from an entire year. During the training process, to maximize the use of data, each episode randomly selects an initial time step from the training set and then employs a fixed one-week warm-up to reset the environment and capture the initial state. Optimization was performed on an hourly basis over one week for each episode. The optimization time step for the entire process was set to one hour. We utilized four standard testing scenarios provided in BOPTEST to evaluate model performance. Descriptions of these scenarios are as follows:

- Peak Heat Day: The testing period comprises a two-week assessment that incorporates a one-week warm-up phase using baseline control. This fortnight is strategically positioned around the day featuring the highest 15-minute system heating load annually.
- Typical Heat Day: The evaluation period constitutes a two-week test, preceded by a one-week warm-up phase employing baseline control. The two-week interval is anchored around the day featuring the maximum quarter-hourly system heating load, which is nearest yet lower than the median of all annual highest quarter-hourly heating loads.
- Peak Cool Day: This study entails a testing period of two weeks, which includes a one-week preliminary phase using baseline control. The two-week duration is strategically aligned to encompass the day of the highest quarterly-hour system cooling load throughout the year.
- Typical Cool Day: This experiment encompasses a two-week testing interval preceded by a one-week acclimation phase, implementing baseline control. The two-week span is aligned with the day that features the maximum 15-minute system cooling load closest, but not exceeding, the median of all daily 15-minute peak cooling loads throughout the year.

Table 4
The hyperparameters of three SAC agents.

Hyperparameter	SAC for M.1	SAC for M.2	SAC for M.3
Learning rate	0.0002	0.0002	0.0002
Discount factor	0.99	0.99	0.99
Soft update	0.005	0.005	0.005
Memory size	50 000	50 000	50 000
Minibatch sampling size	64	64	64
Training steps	168000	168000	168000
Network architecture	Linear: 400 * 300	Linear 400 * 300	GRU: 1 layer; 32 hidden size

All algorithms were implemented using PyTorch and accelerated using an RTX 4090 graphics card. The environmental code was written using the OpenAI Gym framework [55], and the Soft Actor–Critic (SAC) algorithm was trained and tested using the Stable Baselines 3 open-source library [56]. To ensure a fair comparison, we fixed the random seed in all the training and testing processes, ensuring that our experiments could be consistently replicated on any machine.

The final hyperparameters for the SAC are summarized in Table 4, which were determined through manual experimentation. We also attempted to use Optuna, a Python library, for automatic hyperparameter optimization [57]. However, the lack of parallel computational support in BOPTEST-Gym results in excessively long adjustment times. Thus, we could not successfully implement it for our task, nor could we find an optimal global configuration for the hyperparameters. However, because our study compares the effects of different prediction information usages on RL, maintaining consistency in the hyperparameters is sufficient.

The learning rate (0.0002) was chosen based on multiple experiments and experiences. In addition to the learning rate, the hyperparameter settings are fundamentally based on the stable-baseline library as they optimize the hyperparameters, endowing them with improved generalization.

4. Results and discussion

In reinforcement learning problems, we typically focus on improving and evolving the agent's performance during training. Unlike in supervised learning, we are less concerned with variations in the loss function. Figs. 6 and 7 depict the changes in cumulative rewards for different agents throughout their training.

All models were trained for 1000 episodes in the BOPTEST environment. Given that each episode consisted of 168 time steps (corresponding to 168 h per week), the total number of time steps for all episodes was 168,000. After a certain training period, all sessions converged

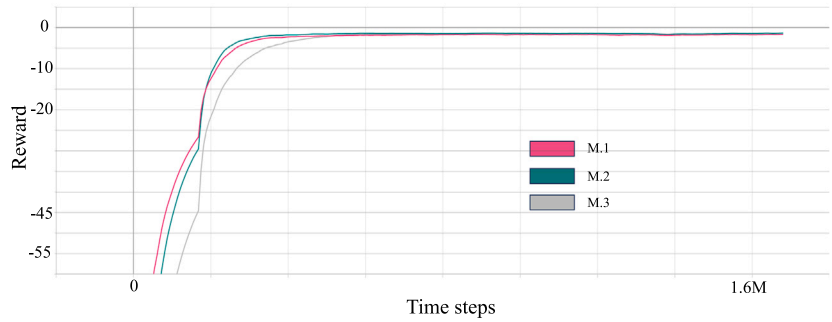

Fig. 6. Evolution of average return per episode of three SAC agents in cost and thermal comfort.

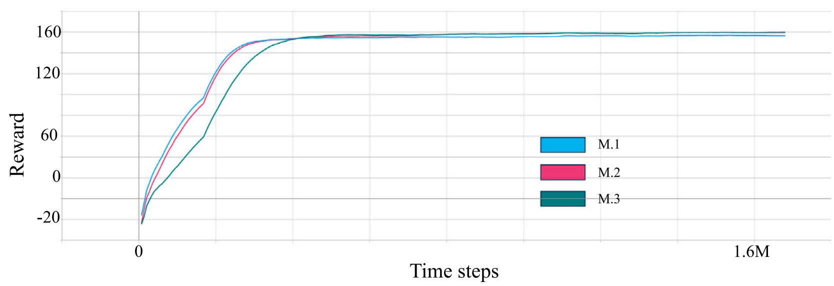

Fig. 7. Evolution of average return per episode of three SAC agents in thermostatic control.

Table 5
Performance results of different controllers during the heating periods under the dynamic pricing scenario.

	Peak heat			Typical heat		
	M.1	M.2	M.3	M.1	M.2	M.3
Operational cost ($\$/m^2$)	0.045	0.038	0.036	0.049	0.044	0.039
Indoor thermal comfort index	0.253	0	0	0	0.061	0.033

Table 6
Performance results of different controllers during the cooling periods under the dynamic pricing scenario.

	Peak cool			Typical cool		
	M.1	M.2	M.3	M.1	M.2	M.3
Operational cost ($\$/m^2$)	0.099	0.094	0.092	0.063	0.055	0.052
Indoor thermal comfort index	0.029	0.005	0.080	1.758	0.122	0.204

to a relatively stable form. The training curves were consistent with those of general RL algorithms. Notably, regardless of the task reward function, M.1 converged at the fastest rate, whereas M.3 converged at the slowest. This slowdown was expected because a GRU neural network was used in M.3. The GRU network processes sequences based on temporal variations, which precludes the possibility of rapid, parallel computations on a large scale. Additionally, the network requires sequential processing of the state vectors to capture the characteristics of time-series changes as effectively as possible. From the perspective of the final converging results, it appears that the convergence speed has not been significantly impacted.

4.1. Statistic experiments in four testing episodes

4.1.1. Reward based on user comfort and running costs

The overall performance indicators of the four test episodes are listed in Tables 5 and 6. The Operational Cost and the Indoor Thermal Comfort Index were calculated within the BOPTEST framework. The Indoor Thermal Comfort Index indicates the extent to which the system violates the upper and lower bounds of comfort during operation. Higher values indicate poorer performance in terms of thermal comfort.

The results in Table 5 suggest that the reinforcement learning model employing predictive information outperformed Model M.1, which did not utilize a predictive approach, in both heating cycles. When tests were conducted during the peak heat period, the M.2 device, with added predictive information, reduced operational costs by 15.6%. The M.3 model, which employs a GRU network for handling predictive information, is capable of reducing operational costs by 20%, marking a 5 percentage point improvement over conventional networks. In terms of indoor thermal comfort, both M.2 and M.3 utilized predictive information and successfully maintained the indoor temperature within the comfort range for 168 h. When the experimental case shifts to a typical heat scenario, the M.2 model, which solely uses predictive information, can reduce the operational costs by 10.2%. The M.3 model, which uses a GRU network, can decrease expenses by 20.4%, which is the best performance among the three models; this confirmed the effectiveness of employing a GRU network for processing predictive data. In terms of indoor thermal comfort, Model 1 (M.1) exhibited the best performance. Model 3 (M.3), with the integration of a GRU network, exhibited a comfort level improvement of 46% over Model 2 (M.2). Given that our reward function primarily emphasizes operating cost considerations, these results are reasonable and expected.

The cooling cycle presented in Table 6 corroborates the conclusions of previous studies. M.3 consistently achieved the best performance in terms of operational costs. In peak cooling scenarios, M.3 saves 7% of the operational costs compared to M.1; in typical cooling scenarios, the savings increase to 17.5%. In terms of indoor temperature control, M.3 performed the worst during all cooling periods. This showed an 88.4% improvement over Model M.1, which did not utilize predictive information. However, given that we assigned a 16-fold weight to the operational costs of the reward function, this performance is reasonable. This demonstrates the efficacy of the GRU model in handling predictive information for reinforcement learning and current tasks.

In this context, we observe that during peak cooling periods, Model M.3 does not maintain optimal indoor thermal comfort, underperforming compared to Models M.1 and M.2, and particularly against M.1.

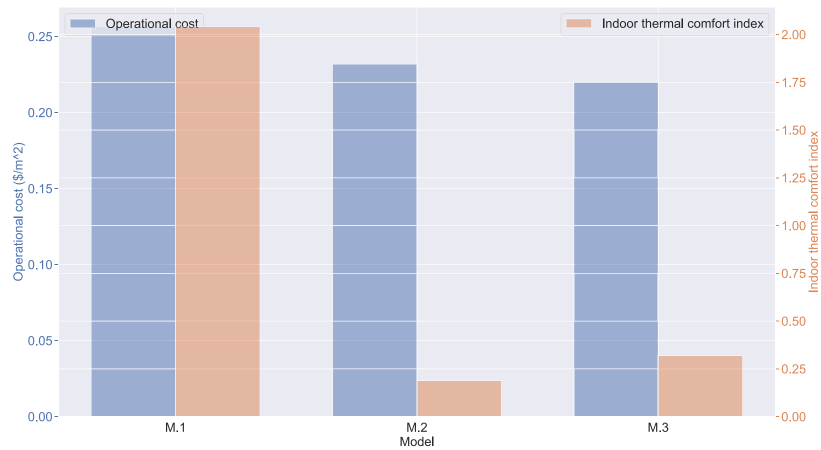

Fig. 8. Comparison of overall operational results based on thermal comfort and running costs.

Table 7
Performance results of different controllers under the dynamic pricing scenario and thermostatic control.

	Average temperature difference per episode [°C]		
	M.1	M.2	M.3
Peak heat	0.369	0.322	0.283
Typical heat	0.309	0.309	0.232
Peak cool	0.414	0.287	0.231
Typical cool	0.390	0.243	0.238

This disparity is likely attributable to the configuration of the reward function. In experiments balancing operational costs and thermal comfort, we assigned a 16-fold greater weight to the reward for operational cost efficiency. Since reinforcement learning aims to maximize the overall reward, this naturally led to a greater focus on operational costs. The rationale behind this 16-fold weighting was primarily to achieve a numerical balance. Within the BOPTEST framework, the absolute numerical value of operational costs is lower than that of the thermal comfort index. Consequently, it became necessary to appropriately balance the rewards for these two objectives. This equilibrium in reward allocation posed no issues in the initial stages of training. However, as training progressed, the calculated numerical value of indoor thermal comfort tended to decrease. This decline resulted in the DRL algorithm increasingly favoring the operational cost objective in the later stages of training, thereby leading to the observed outcomes.

Finally, we computed the sum of the performance metrics for different models across the four scenarios and plotted the results in bar graphs for ease of overall comparison (Fig. 8).

4.1.2. Reward based on thermostatic control

The overall performance indicators of the four test episodes are listed in Table 7. At this point, the optimization objective of RL is a univariate function. At this point, RL focuses solely on maintaining a constant room temperature without introducing other optimization objectives that could distract from RL's performance. The term "Average Temperature Difference" denotes the average magnitude of the absolute difference between the indoor temperature and the target temperature of 22 degrees Celsius across all time steps in an episode.

The performances of models M.2 and M.3 surpass that of model M.1, which lacks predictive data; this is consistent with our understanding of the benefits of incorporating predictive data, particularly for delicate tasks such as temperature regulation. Likewise, M.3 consistently outperformed M.2 across all scenarios, demonstrating the enhanced predictive capability of the GRU networks. In the peak cooling scenario, the thermal control index of M.3 improved by approximately 44.2% compared to M.1. This is the most significant improvement among the four scenarios. Furthermore, a typical heat scenario is worth noting. The performance of M.2 and M.1 remained consistent, suggesting that the mere addition of predictive information did not enhance the model's performance in this case. However, the performance of M. 3 improved by approximately 24.9% when using the GRU network; this substantiates the necessity of employing the GRU network. Experiments focusing on a single optimization objective and reward function have validated the effectiveness of the GRU-RL approach. This further illustrates that the suboptimal results observed in previous experiments were due to an imbalance in the setting of the reward function.

4.2. Detail visualization analysis

In addition to statistical data, the authors aimed to illustrate specific strategies and temperature variations in practical operations, thereby enabling readers better understand the control effects of RL under different tasks. Given the length constraints, the authors opted for an in-depth analysis of typical heating and cooling scenarios in terms of user comfort and running costs. As the authors will make the model's training outcomes and testing environments open-source, interested readers can download them to test various scenarios independently.

Figs. 9 and 10 respectively depict the detailed temperature variations controlled by different intelligent agent controllers in typical heating and cooling scenarios. The dashed lines in the figures represent the range of indoor comfort bounds throughout the cycle. The range of comfortable temperature limits is narrower, between 21 and 24 degrees Celsius when there is human activity indoors. In contrast, without such activity, the range widens, stretching from 15 to 30 degrees Celsius.

From the operational control results, the models utilizing predictive information saved more energy than Model M.1 during periods when the building was unoccupied. For instance, at time step 125 in Fig. 9, the indoor temperatures of Models M.2 and M.3 were lower than those of Model M.1. During this time, the building is unoccupied, and air conditioning can be turned off while maintaining the indoor temperatures within the comfort zone. In certain instances, M.3 outperformed M.2, as demonstrated by the 50th time step in Fig. 9, where M.3 displays a lower temperature, thus resulting in more energy savings.

Analogous conclusions were drawn for the typical cooling scenarios. For instance, at Step 25 in Fig. 10, both M.3 and M.2 have learned similar pre-cooling controls. However, M.3 applied a gentler pre-cooling approach, avoiding excessive temperature reduction, thereby reducing operational costs. Models M. 2 and M. 3, which incorporate predictive information, learned to perform pre-cooling operations similar to those that the M.1 model failed to grasp in typical cool situations. This finding underscores the importance of using predictive data. The slightly superior performance of M.3 reflects the effectiveness of the GRU network in processing the observations.

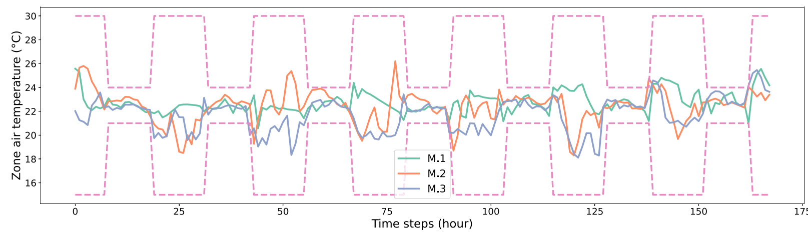

Fig. 9. Zone air temperature changing in typical heat.

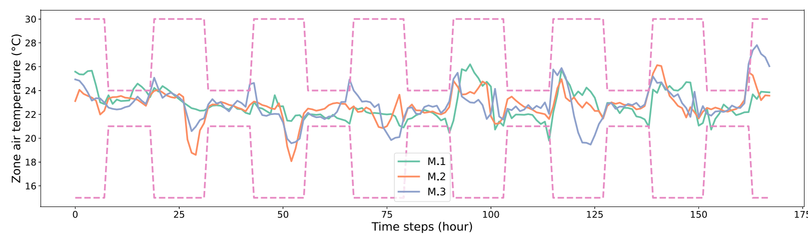

Fig. 10. Zone air temperature changing in typical cool.

Finally, the authors emphasized the impact of incorporating predictive information. During heating cycles, this information can be used to appropriately turn off the air conditioning during building downtime and preheat ahead of use, effectively saving energy (as the operational costs during non-peak times are lower). The cooling process also allows suitable precooling operations to be learned. Moreover, M.3 proved to be more reasonable than M.2 during precooling and preheating, resulting in significant cost savings. This further improvement was due to the advantages of the GRU network over pure prediction.

5. Conclusion, limitation, and future work

RL shows significant potential for addressing the crucial demand for reducing the energy use and carbon footprint of HVAC systems while ensuring indoor comfort. The authors developed an RL algorithm that uses GRU networks to handle predictive data based on the open-source framework BOPTEST to better address the issue of incorporating predictive information into RL algorithms. The experimental objectives included the joint control of indoor thermal comfort and operational costs, as well as the maintenance of indoor temperature stability. The proposed method was compared with models that did not use predictive information and those that processed predictive information with regular networks. Optimization tests were conducted under four built-in heating and cooling scenarios of BOPTEST, with the following key findings:

- The proposed GRU-RL combined model achieves the best overall operational costs and thermal comfort index across four scenarios when the operational objectives are indoor thermal comfort and cost-efficiency. The overall operational cost was reduced by 14.5% compared to the model that did not utilize predictive information, while the indoor thermal comfort performance improved by 88.4%. While the performance of the GRU-RL model may be inferior to models using standard networks in terms of indoor comfort, this result is justifiable considering the reduction in operational costs and the significant computational weightage of operational costs in the reward function.
- In the context of indoor temperature control tasks, we solely utilize a single optimization objective to better compare the effects of incorporating predictive information and the use of GRU models. Compared to models that do not utilize predictive information, the GRU-RL model enhances the performance of thermostatic control by approximately 14.2%. Furthermore, it outperforms models that use ordinary networks to incorporate predictive information, showing an improvement of around 5%. The GRU-RL model demonstrated superior performance in all four scenarios, validating the efficacy of incorporating the GRU model.
- Through detailed analysis of two scenarios, we found that the introduction of predictive information enables the model to learn pre-cooling and pre-heating actions, thereby controlling both indoor comfort and operating costs. The model will appropriately turn off the air conditioning during unoccupied periods and initiate pre-cooling and pre-heating in advance. The GRU-RL model's operations are smoother during this pre-cooling and pre-heating process, further reducing operating expenses.
- To tackle the challenge of reproducibility in our research, we have developed a test platform grounded in the BOPTEST open-source framework. Additionally, we have ensured that all algorithms employed in our study are open-source, thereby facilitating independent replication and validation of our findings by the broader research community. Our integration of the BOPTEST framework with reinforcement learning algorithms is also readily accessible, providing a valuable reference and tool for future researchers in this domain.

However, because of the lack of parallel computing support in BOPTEST, we were unable to select optimal hyperparameters. Furthermore, a detailed discussion of the setting of the reward function weights is precluded owing to computational time constraints. Future research will apply this algorithm to more complex scenarios, develop fixed and efficient algorithms for the design of reward functions, and replace the GRU model with other models to enhance the interpretability of RL. Moreover, in the experiments balancing operational costs and indoor thermal comfort, we demonstrated that the design of the reward function in reinforcement learning significantly affects the optimization results. Therefore, a crucial future research direction will be how to effectively and automatically design adaptive reward functions for multi-objective optimization.

CRediT authorship contribution statement

Yuan Gao: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **Shanrui Shi:** Validation, Visualization, Writing – original draft, Writing – review & editing. **Shohei Miyata:** Funding acquisition, Project administration. **Yasunori Akashi:** Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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